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Summary

Computational and physical organic chemist with extensive experience in reaction mechanism analysis, catalytic systems, stereoselectivity, and molecular modeling using DFT, QM/MM, and molecular dynamics approaches. Experienced in interpreting complex chemical problems through mechanistic reasoning, kinetic isotope effects, wavefunction analysis, and computational workflows. Skilled in Python-based scientific scripting, computational chemistry automation, and AI/ML-assisted chemistry analysis. Proven ability to communicate advanced chemistry concepts clearly through structured scientific explanations, visual interpretation, and collaborative interdisciplinary research.

AI/ML & Cheminformatics:

Currently expanding expertise in AI/ML applications in chemistry through targeted training programs, including hands-on Data Science (Codegnan Institute, Hyderabad), AI/ML in Chemistry (IIT Madras-SWAYAM Plus), Cheminformatics-focused ML (BioTecNika Campus), and with emphasis on Python-based modeling, data analysis, and predictive methods.

	Organization	Research	Supervisor	Year
Postdoctoral Associate				
1.	Ewha Womans University, Seoul, South Korea.	Brain Pool Fellowship	Sun Choi	24/07/2026-present
2.	IISER, TVM, India	C-C coupling reactions via Photo - Radical mediated in Organic Synthesis	Y. Veera Reddy	21/01/2026-10/07/2026
3.	Southern Methodist University, TX, USA.	Studies of Homogeneous and Enzyme Catalysis	Elfi Kraka	01/12/2022-08/15/2025
4.	State University of New York, NY, USA	Studies of Homogeneous Catalysis – Kinetic Isotopic Studies	Mathew J. Vetticatt	15/08/2022-30/11/2022-
5.	University of Pennsylvania, PA, USA.	C-C Coupling reactions using QM studies	Marisa C. Kozlowski	15/08/2020-15/08/2022-

6.	UNIST, Ulsan, South Korea.	Small Molecules using QM and ab initio Studies	Kwang Soo Kim	01/10/2018-31/07/2020-
Doctoral Studies (Ph. D) in Chemistry				
IIT Bombay, India		Computational Studies of Asymmetric Dual Chiral Catalysis	R. B. Sunoj	2012-2018
Teaching and Prior Research Experience				
NIT, Kurukshetra, India		Research Assistant	--	2010-2012
SVIT, Hyderabad, India		General Chemistry (Teaching)	--	2009-2010
Masters in Chemistry (M. Sc)				
Andhra University, India		Analytical Chemistry	--	2005-2007
Bachelors in Chemistry (B. Sc)				
Government Degree College, Tuni, Andhra University, India		Mathematics, Physics, Chemistry	--	2002-2005

Publications

1. **Bhaskararao, B**.; Antonio, J. J.; Kraka, E. Umpolung Character of Styrenes: Mechanism and Stereoselective Studies of α, β-substituted Amino Acids using Chiral Ir-Phosphine Complexes. *J. Comput. Chem.* **2026,** 47, e70477. http://dx.doi.org/10.1002/jcc.70477
2. Basnet, P.; **Bhaskararao**, **B**.; Amadeo; P. A.; Hendrick; C. E.; VanGelder; K. F.; Curto, J. M.; Kim, D. Y.; Hong, G.; Kozlowski. M. C. Heterogeneous Palladium(0) Catalysts for C(sp3)–H Activation and Coupling to Form Hindered C–C Bonds. *ACS Catal.* **2026**. https://doi.org/10.1021/acscatal.6c04199
3. Maji, S.; Maiti, S.; Anusha, E.; Guin, S.; **Bhaskararao, B**.; Maiti, D. Ligand-Controlled Pd(II)-Catalyzed α,β-Unsaturation and β-Arylation of Long-Chain Acids. *J. Am. Chem. Soc.* **2026,** *148*, 21558-21571. https://doi.org/10.1021/jacs.6c00988
4. Bisoyi, A.; Murgeshan, R.; K G, G.; Tripathy, A. R.; Behera, A.; **Bhaskararao, B**.; Yatham, V. R.; Photoinduced Alkynylation of Alkylboronic Acids and Alkyl-Boronates via Aminyl Radical Transfer (ART) Strategy. *Chem. –A Eur. J.* **2026**. e71121. https://doi.org/10.1002/chem.71121
5. Joshi, C.; Dorsheimer, J. R.; Nyagilo, V. O.; Ramos, M. C; Banerjee, S.; **Bhaskararao, B**.; Shen, Y.; Rovis, T.; Vetticatt. M. J. Probing the Oxidative Addition Transition State in

Nickel Metallaphotoredox Catalysis using Intramolecular ^{13}C Kinetic Isotope Effects. *J. Am. Chem. Soc.* **2025,** *147*, 24275–24285. https://pubs.acs.org/doi/10.1021/jacs.4c10182

6. Volchkov, I.; Powell, B. V.; Zatolochnaya, O. V.; Leung, J. C.; Pennino, S.; Wu, L.; Gonnella, N. C.; **Bhaskararao, B.**; Kozlowski, M. C.; Reeves, J. T. A Practical Synthesis of Terminal Vinyl Fluorides. *J. Org. Chem.* **2023**, *88*, 10881–10904. https://doi.org/10.1021/acs.joc.3c00917
7. Kajal, S.; Jeong, J.; Seo, J.; Anand, R.; Kim, Y.; **Bhaskararao, B.**; Beom Park, C.; Yeop, J.; Hagdfeldt, A.; Young Kim, J.; Kim, K. S. Coordination Modulated Passivation for Stable Organic-Inorganic Perovskite Solar Cells. *Chem. Eng. J.* **2023**, *451*, No. 138740. https://doi.org/10.1021/jacs.2c07376
8. **Bhaskararao, B.**; Rotella, M. E.; Kim, D. -Y.; Kee, J.-M.; Kim, K. -S.; Kozlowski. M.C. Ir and NHC Dual Chiral Synergetic Catalysis: Mechanism and Stereoselectivity in γ-Butyrolactone Formation. *J. Am. Chem. Soc.* **2022**, *144*, 16171–16183. https://doi.org/10.1016/j.cej.2022.138740
9. Amadeo, P.; **Bhaskararao, B.**; Yang, Y.-F.; Kozlowski, M.C. Inherent Selectivity of Pd C–H Activation from Different Metal Oxidation States. *Organometallics* **2021**, *40*, 2290–2294. https://doi.org/10.1021/acs.organomet.1c00011
10. **Bhaskararao, B.**; Kim, D. Y.; Madridejos, M. L. J.; Park, C. –M.; Kim, K. S. Rational Design of Metal-Ligands for Conversion of CH_4 and CO_2 to Acetates: Role of Acids and Lewis acids. *J. Mater. Chem. A,* **2020**, *8*, 14671-14679. https://doi.org/10.1039/D0TA04002A
11. Changotra, A.; **Bhaskararao, B.**; Hadad, C. M.; Sunoj, R. B. Insights on Absolute and Relative Stereocontrol in Stereodivergent Cooperative Catalysis. *J. Am. Chem. Soc.,* **2020**, *142*, 9612-9624. https://doi.org/10.1021/jacs.9b13962
12. Singh, S.; Pareek, M.; Changotra, A.; Banerjee, S.; **Bhaskararao, B.**; Balamurugan, Palaniappan.; Sunoj, R. B. A Unified Machine Learning Protocol for Asymmetric Catalysis: A Proof of Concept Demonstration using Asymmetric Hydrogenation. *Proc. Natl. Acad. Sci.* U. S. A. **2020**, *117*, 1339. https://doi.org/10.1073/pnas.1916392117
13. **Bhaskararao, B.**; Singh, S.; Anand, M.; Verma, P.; Prakash, P.; Athira C, Malakar, S.; Schaefer, H. F.; Sunoj, R. B. Is silver a mere terminal oxidant in palladium catalyzed C–H bond activation reactions? *Chem. Sci.,* **2020**, *11*, 208. https://doi.org/10.1039/C9SC04540F
14. **Bhaskararao, B.**; Sunoj, R. B. Two chiral catalysts in action: insights into cooperativity and stereoselectivity in proline and cinchona-thiourea dual organocatalysis. *Chem. Sci.,* **2018**, *9*, 8738. https://doi.org/10.1039/C8SC03078B
15. Deb, A.; Singh, S.; Seth, K.; Pimparkar, S.; **Bhaskararao, B.**; Guin, S.; Sunoj, R. B.; Maiti, D. Experimental and Computational Studies on Remote γ-C(sp3)–H Silylation and

Germanylation of Aliphatic Carboxamides. *ACS Catal.* **2017**, *7*, 8171-8175. https://doi.org/10.1021/acscatal.7b03056

16. Dutta, U.; Modak, A.; **Bhaskararao, B.**; Bera, M.; Bag, S.; Mondal, A.; Lupton, D. W.; Sunoj, R. B.; Maiti, D. Catalytic Arene Meta-C–H Functionalization Exploiting a Quinoline-Based Template. *ACS Catal.* **2017**, *7*, 3162-3168. https://doi.org/10.1021/acscatal.7b00247
17. **Bhaskararao, B**.; Jindal, G.; Sunoj, R. B. Exploring the Mechanism and Stereoselectivity in Chiral Cinchona Catalyzed Hetero-dimerization of Ketenes. *J. Org. Chem.* **2017**, *82*, 13449. https://doi.org/10.1021/acs.joc.7b02517
18. **Bhaskararao, B.**; Sunoj, R. B. Asymmetric Dual Chiral Catalysis us ing Iridium Phosphoramidites and Diarylprolinol Silyl Ethers: Insights into Stereodivergence. *ACS Catal.* **2017**, *7*, 6675–6685. https://doi.org/10.1021/acscatal.7b02776
19. Maji, A.; **Bhaskararao, B.**; Singha, S.; Sunoj, R. B.; Maiti D. Directing Group Assisted Meta-Hydroxylation by C–H activation. *Chem. Sci.,* **2016**, *7*, 3147. https://doi.org/10.1039/C5SC04060D
20. **Bhaskararao, B**.; Sunoj, R. B. Origin of Stereodivergence in Cooperative Asymmetric Catalysis with Simultaneous Involvement of Two Chiral Catalysts. *J. Am. Chem. Soc.* **2015**, *137*, 15712 –15722. https://doi.org/10.1021/jacs.5b05902

Computational Tools & Software Proficiency

- **Quantum Chemistry**:
  Gaussian– for DFT and QM/MM calculations including reaction mechanism and stereoselectivity studies and other supporting AIM, NBO, LMA, BSO and URVA studies.
  VASP – for ab initio periodic DFT simulations of solid-state materials.
- **Molecular Dynamics**:
  AMBER, GROMACS – for classical MD simulations, force field parametrization, and free energy calculations and Schrodinger - Maestro for Molecular docking.
- **Programming & Scripting**:
  Python – data analysis, automation of computational workflows, plotting, and molecular descriptor generation.
  Shell/Bash scripting – automation of quantum chemistry and MD job submission, file processing, and high-performance computing (HPC) workflow management.

As a Active Reviewer

- 50 and more Research articles for the journals mentioned below were appraised by me as a "Reviewer" including **Nature Catalysis**, **Nanoscale**, **The Journal of Physical Chemistry**, **Organic Chemistry Frontiers**, **Inorganic Chemistry** (IC), **Computational and Theoretical Chemistry** (CTC), **JOC**, New Journal of Chemistry, Catalysts, Symmetry, Energies, the international journal of molecular sciences (IJMS).